

HASSAN USMAN KATSINA POLYTECHNIC

DEPARTMENT OF ELECTRICAL ENGINEERING

PROJECT PROPOSAL

**Design and Implementation of a Smart USB Power Socket with Android
Monitoring Application**

PRESENTED BY:

Yusuf Umar Saleh

Matric Number: N23EE010

Session: 2025 / 2026

SUPERVISOR:

[SUPERVISOR NAME]

1 INTRODUCTION

1.1 Background to the Study

The Universal Serial Bus (USB) has evolved from a peripheral-interconnection standard into the dominant power-delivery and charging interface for portable consumer electronics, embedded systems, and Internet-of-Things (IoT) sensors. According to the USB Implementers Forum (2024), more than seven billion USB-enabled devices shipped globally in 2024, spanning smartphones, tablets, wearable gadgets, Bluetooth headsets, and low-power IoT nodes. In Nigeria, the rapid penetration of affordable Android smartphones and the expansion of mobile broadband have made the USB socket a near-universal feature of vehicular interiors, household extension boards, solar home systems, and small-office power distribution units (Nigerian Communications Commission, 2024).

Despite this ubiquity, the majority of USB sockets in the Nigerian market remain passive power sources. They provide a regulated 5 V DC output but offer no visibility into voltage stability, current draw, energy consumption, or operational state. This opacity creates operational and economic problems for users. There is no early warning of regulator drift or component ageing, no mechanism to schedule charging around off-peak electricity tariffs, and no capability to remotely interrupt charging once a device is fully replenished. These limitations result in energy waste, reduced battery lifespan, and prevent demand-side management of USB-powered assets (Adewale, Okafor, & Onuoha, 2023).

The convergence of low-cost Wi-Fi microcontrollers, such as the Espressif ESP8266, with AI-assisted mobile development frameworks, such as Google Antigravity, has created a realistic pathway to retrofit intelligence into legacy USB power sockets. When such sockets are paired with Android applications, even engineering students can build production-quality monitoring applications within a single academic session (Google, 2025). The proposed work therefore seeks to design and implement a Smart USB Power Socket that combines a proven linear-regulator circuit built around the LM317 with an ESP8266-based Wi-Fi telemetry stage and an Android monitoring application developed using Google Antigravity.

1.2 Statement of the Research Problem

Conventional USB power sockets available in the Nigerian market convert a 12 V battery or mains-derived DC source to a nominal 5 V output using discrete linear regulators such as the LM317 or its fixed-voltage counterparts. While these circuits are inexpensive and well-understood, they exhibit four operational limitations that the proposed smart socket seeks to address.

First, there is an absence of remote monitoring. Users cannot, from a phone or computer, ascertain whether the socket is delivering the expected 5 V, how much current a connected device is drawing, or whether the regulator is operating within its safe thermal envelope. This lack of visibility contributes to premature device failure and makes root-cause analysis difficult when a connected smartphone charges slowly or refuses to charge at all. Second, there is an absence of remote control; once a device is plugged into a conventional USB socket, the only way to interrupt charging is to physically unplug it, which is inconvenient in scenarios such as

overnight charging, scheduled charging of battery banks, or remote power-cycling of IoT sensors deployed in hard-to-reach locations.

Third, there is an absence of energy accounting. Without measuring current and voltage over time, it is impossible to compute watt-hours consumed, which in turn makes it impossible to estimate the cost of charging or to identify inefficient chargers and faulty cables. In a country where off-grid solar installations are increasingly common, such information is critical for sizing battery banks and solar panels. Fourth, the absence of a scheduling mechanism means that USB sockets cannot participate in demand-side management strategies, depriving users of the tangible economic benefits of deferring charging to low-tariff windows during grid load-shedding (Adeyemi, Bello, & Okoro, 2021).

1.3 Justification of the Study

The proposed study is justified on four grounds. First, the study addresses a real and currently unmet need in the Nigerian consumer-electronics landscape, where the vast majority of USB sockets in use remain passive and unmonitored. The proposed smart socket provides voltage stability monitoring, current measurement, energy accounting, scheduling, and remote control within a single integrated device, addressing all four limitations identified in Section 2.2.

Second, the study is timely. The constituent technologies—the ESP8266 Wi-Fi microcontroller, the LM317 regulator, and AI-assisted Android development using Google Antigravity—are now mature and affordable enough that a complete smart socket can be built for under fifty thousand Naira, well within the budget envelope of a Nigerian project and competitive with imported alternatives. Third, the study is original in the Nigerian academic context, as no published work in the local literature has yet documented the integration of a classical linear-regulator design with the ESP8266 and a Google Antigravity-built Android application. Fourth, the study is feasible within the timeframe of a single academic session; all required components are available from Nigerian online stores such as Jumia, Konga, Microscale, and Nicrobit, and the firmware and software development toolchains are free and well-documented.

1.4 Research Aim and Objectives

The main aim of this project is to design and implement a Smart USB Power Socket with Android monitoring and control capability, suitable for Nigerian residential, automotive, and small-office applications. The specific objectives through which this aim will be achieved are to:

- i. Design and construct a 12 V-to-5 V DC-DC conversion stage using the LM317 adjustable linear regulator, with appropriate reverse-polarity protection, input and output filtering, over-voltage clamping, and power indication.
- ii. Interface an ESP8266 NodeMCU Wi-Fi module to the regulator output through a voltage divider and an ACS712 hall-effect current sensor, and develop firmware that publishes real-time voltage, current, and power readings over the local Wi-Fi network.
- iii. Implement an Android monitoring application using Google Antigravity as the AI-assisted development environment, providing remote ON/OFF control, a real-time and

historical power-consumption chart, and a scheduling module for time-based automation.

- iv. Integrate a relay-based galvanic isolation stage between the regulator output and the USB Type-A socket to enable safe remote control of charging.
- v. Test and evaluate the complete system for voltage stability across load currents of 0–1.2 A, app response time across Wi-Fi distances of 1–15 m, and energy-metering accuracy against a calibrated benchtop multimeter.

1.5 Significance of the Study

The proposed study is significant in several dimensions. From an academic standpoint, it demonstrates the integration of classical analogue electronics (the LM317 regulator family) with modern embedded Wi-Fi (ESP8266) and AI-assisted mobile development (Google Antigravity) within a single academic deliverable. It therefore serves as a reference template for future final-year projects that seek to combine hardware, firmware, and software into a coherent IoT system. From an industrial standpoint, the smart socket addresses real needs in the Nigerian context: for households, it provides visibility into the cost of charging mobile devices and supports demand-side management; for small businesses such as phone-repair workshops, cybercafés, and charging kiosks, it enables energy accounting and asset tracking; for automotive applications, it allows drivers to monitor the state of in-vehicle USB charging ports without manual inspection.

From a research standpoint, the system provides a low-cost experimental platform for studying load profiles of USB-powered devices, regulator thermal behaviour under continuous duty, and the trade-offs between linear and switch-mode regulation in cost-sensitive applications. The hardware bill of materials is intentionally kept affordable to enable replication by other Nigerian engineering researchers. Finally, the project contributes to the localisation of smart-home technology in Nigeria: the vast majority of commercial smart sockets sold locally are imported, costly, and poorly supported. By demonstrating that a functionally equivalent device can be designed, fabricated, and programmed locally, this work contributes to the broader agenda of building indigenous capacity in IoT engineering.

2.7 Project Deliverables and 3D Working Model

In addition to the physical hardware, firmware, and Android application described in this proposal, the project includes an interactive three-dimensional (3D) working model of the complete smart-socket system, deployed as a web application. The 3D model was developed using React Three Fiber (a React renderer for Three.js) and Next.js, and is hosted on Vercel for public access. It visualises every component of the system—the 12V battery, 1N4007 diode, LM317 voltage regulator, filter capacitors and Zener diode, USB Type-A socket, ACS712 current sensor, voltage divider, ESP8266 NodeMCU, 5V relay module, Wi-Fi router, and the Android device running the Google Antigravity application—as stylised 3D meshes connected by colour-coded wires that animate the flow of current and data when the system is powered on.

The 3D model serves three pedagogical and presentation purposes. First, it provides supervisors, examiners, and fellow students with an intuitive, interactive visualisation of the system architecture that complements the schematic and block diagrams presented later in this proposal. Second, it allows users to click on any individual component to view its full

specification, role in the system, and list of connections to other components, thereby functioning as a self-documenting reference. Third, it includes a signal-path filter that isolates each of the four functional paths (power, sensing, control, and communication), making it easier to understand how each subsystem contributes to the overall operation of the smart socket.

The 3D working model is publicly accessible at: <https://smart-usb-power-socket-3d.vercel.app/>.

The web application is fully responsive and works on desktop computers, tablets, and smartphones without requiring any installation. It loads in any modern web browser (Chrome, Firefox, Edge, Safari) and does not require a login. The model will be referenced live during the proposal defence and the final project presentation, and will remain accessible after the project is completed as a permanent digital companion to the physical hardware deliverable.

2. MATERIALS AND METHODS

This section presents the techniques, materials, and equipment that will be used to design, construct, and evaluate the Smart USB Power Socket. The methodology follows a top-down approach: the system is first described at the block-diagram level, then each functional block is elaborated in turn, and finally the firmware and software layers that bind the hardware into a coherent smart-socket system are presented.

2.1 System Architecture and Design Approach

The Smart USB Power Socket comprises five functional blocks: (i) the power input and protection block, (ii) the voltage regulation block, (iii) the sensing block, (iv) the control block, and (v) the communication block. The power input and protection block accepts 12 V DC from a cigar-lighter plug or mains-derived adapter and provides reverse-polarity protection through a 1N4007 diode and input filtering through a 470 μ F electrolytic capacitor. The voltage regulation block, centred on the LM317, reduces the 12 V input to a regulated 5 V output suitable for USB devices. The sensing block measures the regulator output voltage through a voltage divider and the load current through an ACS712 hall-effect sensor, both interfaced to the analogue-to-digital converter of the ESP8266. The control block uses a relay placed between the regulator output and the USB Type-A socket to interrupt the load under firmware command. The communication block, implemented by the ESP8266's built-in Wi-Fi radio, hosts a lightweight HTTP endpoint that exposes status JSON and accepts control commands from the Android application. The design approach is modular and staged: each block will be constructed and tested independently before integration, in line with the engineering reproducibility standards articulated by Vincent, Okonkwo, and Bello (2023).

2.2 Circuit Design and Analysis Method

The output voltage of the LM317 regulator is set by an external resistor divider R1 and R2 according to Equation 1, where V_{ref} is the internal 1.25 V reference and I_{adj} is the adjustment-pin current (typically 50 μ A):

$$V_{out} = V_{ref} \times (1 + R2/R1) + I_{adj} \times R2 \quad \dots (1)$$

Substituting $V_{ref} = 1.25 \text{ V}$, $R_1 = 270 \text{ }\Omega$, and $R_2 = 820 \text{ }\Omega$ yields $V_{out} = 1.25 \times (1 + 820/270) + (50 \times 10^{-6}) \times 820 = 5.09 \text{ V}$, which is comfortably within the USB 2.0 safe band of 4.75–5.25 V (Texas Instruments, 2023). The LM317 will be selected in the TO-220 package and bolted to a $25 \times 25 \times 11 \text{ mm}$ aluminium heatsink to manage the worst-case 8.4 W dissipation at full load. A 5.6 V Zener diode across the output provides over-voltage clamping in the event of regulator failure, while input and output capacitors (470 μF and 10 μF respectively) provide ripple rejection and transient response. A 1N4007 diode in series with the input provides reverse-polarity protection, which is essential in automotive applications where the cigar-lighter socket polarity may be inconsistent across vehicle models (STMicroelectronics, 2023).

2.3 Firmware Development Method

The ESP8266 firmware will be developed in the Arduino IDE using the ESP8266 Arduino Core. The firmware will be organised into five functional modules: Wi-Fi connection management, HTTP server, sensor reading, relay control, and the main loop. The Wi-Fi module will use the ESP8266WiFi library to connect the device to the local 2.4 GHz network in station mode and to reconnect automatically in the event of disconnection. The HTTP server, implemented using the ESP8266WebServer library, will expose four endpoints: GET /status returns the current voltage, current, power, and relay state as a JSON object; POST /relay with body {"state": "on"|"off"} commands the relay; GET /history?hours=24 returns an array of timestamped power readings for the chart; and GET /schedules returns the list of stored schedules (Babangida, Idris, & Yusuf, 2024).

The sensor reading module will sample the voltage and current ADC channels at 10 Hz. Each sample will be converted to engineering units using calibration constants stored in non-volatile memory. The instantaneous power will be computed as the product of voltage and current. A 60-sample rolling average (corresponding to 6 seconds of data) will be published to the /status endpoint. Every 60 seconds, the average power over the preceding minute will be multiplied by 1/60 hour to obtain watt-hours, which will be accumulated into the daily energy register. JSON was preferred over binary protocols for ease of debugging and for compatibility with the Retrofit Gson converter on the Android side (Senthilkumar, Anitha, & Prabhu, 2024).

2.4 Android Application Development Method

The Android application will be developed in Android Studio with Google Antigravity as the AI-assisted development environment. The app will target Android 8.0 (API level 26) and above, use Kotlin 1.9, Jetpack Compose for declarative UI construction, and follow the Model-View-ViewModel (MVVM) architecture recommended by Google (2024). Networking will be handled by the Retrofit 2.9 library with the Gson converter, asynchronous operations by Kotlin Coroutines, and persistent storage of schedules by the Room database wrapper over SQLite.

The UI layer will consist of three Compose screens: a Dashboard screen that displays the current voltage, current, and power readings together with a toggle button for the relay; a Power Chart screen that displays a 24-hour bar chart of power consumption using MPAndroidChart 3.1; and a Schedule screen that lists configured ON/OFF schedules and allows the user to add new ones. The ViewModel will expose a `StateFlow<UiState>` that the UI collects using the `collectAsStateWithLifecycle` extension. The Repository will abstract the source of data

(network or local cache) and expose suspend functions for the use cases. Google Antigravity will be used to scaffold the MVVM architecture, generate the Retrofit network layer, draft the Jetpack Compose UI for all three screens, and produce the initial MPAndroidChart configuration; the generated code will then be refined, tested, and integrated with the ESP8266 endpoint (Kumar & Sharma, 2025).

2.5 Testing and Evaluation Method

The complete system will be tested along four dimensions. Voltage-regulation stability will be tested using an electronic load capable of sinking 0–2 A at 5 V, with the load current stepped from 0 A to 1.2 A in 0.1 A increments and the output voltage recorded at each step using a calibrated Fluke 87V benchtop multimeter. The pass criterion is that the output voltage remains within the USB 2.0 safe band of 4.75–5.25 V at all tested load currents.

App response time will be tested by placing the Android phone at distances of 1, 3, 5, 8, 10, 12, and 15 metres from the Wi-Fi router and issuing a /status request from the app, with each measurement repeated ten times and the median value recorded. The pass criterion is a median response time below 100 ms. Energy-metering accuracy will be tested by simultaneously logging the system's reported power consumption and the benchtop multimeter's reported power consumption over a 60-minute period with a 0.5 A constant load, and computing the relative error. End-to-end functional behaviour will be tested by performing ten complete cycles of: open app, verify status, toggle relay off, verify charging stops, toggle relay on, verify charging resumes, schedule a 1-minute off-on cycle, verify schedule executes.

2.6 Materials and Equipment

The complete list of materials and equipment required for the project is presented in the budget (Section 5). Key hardware materials include the LM317 voltage regulator, ESP8266 NodeMCU v3 development board, ACS712 current sensor module, 5 V single-channel relay module, USB Type-A female socket (DIP adapter), 12 V cigar-lighter plug, 12 V DC barrel-jack adapter, an assortment of electrolytic and ceramic capacitors, 1N4007 rectifier diodes, 5.6 V Zener diodes, 5 mm LEDs, an assortment of 1/4 W resistors, an 830-point solderless breadboard, jumper wires, a perfboard, a plastic enclosure, a TO-220 heatsink, and miscellaneous consumables (solder, flux, heat-shrink tubing, mounting hardware).

Software tools required include the Arduino IDE (free, open-source) for ESP8266 firmware development, Android Studio with Google Antigravity (free) for Android application development, the Fluke 87V benchtop multimeter (available in the departmental laboratory) for voltage and current calibration, an electronic load (available in the departmental laboratory) for load testing, and a Wi-Fi router (personally owned) for communication testing. All hardware components have been verified as available from Nigerian online stores including Jumia Nigeria, Konga, Microscale.net, Nicrobit.com.ng, and Nerdshed.com.ng, with prices confirmed in June 2025 as detailed in the budget.

4. RESEARCH TIMELINE

The proposed project will be executed over a period of eight (8) months, spanning the 2025/2026 academic session from September 2025 to April 2026. The timeline is structured to

ensure that each major activity—literature review, hardware design, firmware development, Android development, integration, testing, and report writing—receives adequate attention and overlaps sensibly with related activities. Table 4.1 presents the Gantt-style timeline, with each “X” indicating the months during which the corresponding activity will be actively executed.

Table 4.1: Project research timeline (September 2025 – April 2026)

Activity	Jun '20	Jul '25	Aug '25	Sep '25	Oct '26	NOV '26	Dec '1	Dec '26
Literature review & proposal defence	X	X						
Component sourcing & procurement		X	X					
Voltage regulator circuit design & simulation		X	X					
Hardware prototyping (breadboard)			X	X				
PCB soldering & enclosure assembly				X	X			
ESP8266 firmware development				X	X	X		
Android app development (Google AntigraVity)					X	X	X	
System integration & debugging						X	X	
Testing, calibration & evaluation							X	X
Result analysis & report writing							X	X
Final defence & project submission								X

The timeline allows for some flexibility, particularly in the integration and testing phases, where unexpected technical challenges may require additional debugging time. A two-week buffer will be needed to built and accommodate any slippage in the firmware or Android development phases. The final defence and project submission are scheduled for April 2026, in line with the Hassan Usman Katsina Polytechnic academic calendar for the 2025/2026 session.

5. RESEARCH BUDGET

The detailed budget for the proposed project is presented in Table 5.1. All prices are quoted in Nigerian Naira (₦) and reflect the actual retail prices verified from Nigerian online stores in

June 2025. The principal suppliers consulted include Jumia Nigeria (www.jumia.com.ng), Konga (www.konga.com), Microscale (www.microscale.net), Nicrobit (www.nicrobit.com.ng), and Nerdshed (www.nerdshed.com.ng). Prices include the cost of the components only; delivery charges are not included as they vary by location and order size, but a 10% contingency has been added to the subtotal to cover delivery, price fluctuations, and any items damaged during prototyping.

Table 5.1: Detailed research budget with verified Nigerian Naira prices (June 2025)

S/ N	Component & Specification	Qty	Unit Price	Total Price	Source
1	LM317 voltage regulator (TO-220) — 1.25–37 V, 1.5 A adjustable	1	₦350	₦350	Nicrobit.com.ng
2	1N4007 rectifier diode — 1 A, 1000 V, DO-41 axial	1	₦20	₦20	Microscale.net
3	Electrolytic capacitor 470 μ F / 25 V — Aluminium, 8x12 mm	1	₦200	₦200	Jumia Nigeria
4	Electrolytic capacitor 10 μ F / 25 V — Aluminium electrolytic	1	₦100	₦100	Jumia Nigeria
5	Electrolytic capacitor 1 μ F / 10 V — Aluminium electrolytic	1	₦80	₦80	Jumia Nigeria
6	Ceramic capacitor 0.1 μ F / 50 V — Multilayer ceramic	1	₦50	₦50	Jumia Nigeria
7	Resistor assortment pack (1/4 W, 200 pcs) — Carbon film, 5%, multiple values	1	₦5,000	₦5,000	Konga
8	Zener diode 5.6 V / 1 W — 1N5232B, DO-35 glass	1	₦250	₦250	Jumia Nigeria
9	LED 5 mm red — 5 mm diffused red, 2 V, 20 mA	1	₦50	₦50	Jumia Nigeria
10	USB Type-A female socket (DIP adapter) — USB 2.0 Type-A to DIP 2.54 mm	1	₦500	₦500	Jumia / Konga
11	12 V DC barrel-jack power adapter — 12 V, 2 A, 24 W, 5.5x2.5 mm plug	1	₦3,500	₦3,500	Jumia Nigeria
12	12 V cigar-lighter plug (male, fused) — Standard automotive 12 V	1	₦2,500	₦2,500	Jumia Nigeria
13	ESP8266 NodeMCU v3 (CH340) — ESP-12E Wi-Fi, 4 MB flash	1	₦7,000	₦7,000	Nerdshed.com.ng

S/ N	Component & Specification	Qty	Unit Price	Total Price	Source
14	ACS712 current sensor module (5 A) — Hall-effect, ± 5 A, 66 mV/A	1	₦4,000	₦4,000	Konga
15	5 V relay module (1 channel) — 5 V coil, 10 A, opto-isolated	1	₦1,650	₦1,650	Microscale.net
16	830-point solderless breadboard — MB-102, 830 tie-points	1	₦6,999	₦6,999	Jumia Nigeria
17	Jumper wire pack (3 sets of 40) — Dupont 2.54 mm, 20 cm, M-M/M-F/F-F	1	₦2,500	₦2,500	Jumia Nigeria
18	Perfboard (single-side plated) — 7x9 cm, 2.54 mm pitch	1	₦800	₦800	Jumia Nigeria
19	Plastic project enclosure — 100x60x30 mm ABS	1	₦1,500	₦1,500	Jumia Nigeria
20	Heatsink for TO-220 — 25x25x11 mm aluminium	1	₦500	₦500	Jumia Nigeria
21	Miscellaneous (solder, flux, heat-shrink, screws) — Consumables pack	1	₦2,000	₦2,000	Jumia / local market
	SUBTOTAL			₦39,549	
	Contingency (10% of subtotal)			₦3,955	
	GRAND TOTAL			₦43,504	

The grand total for the complete bill of materials is ₦43,504, inclusive of a 10% contingency. This budget does not include the cost of the Android smartphone used as the controlling device, nor does it include the cost of laboratory equipment such as the benchtop multimeter and electronic load, which are available in the Department of Electrical Engineering laboratory and will be used at no additional cost. The budget is competitive with imported smart sockets of comparable functionality and demonstrates the affordability of locally fabricating IoT devices in Nigeria.

6. REFERENCES

The references below are organised in APA 6th edition style, in alphabetical order by first author surname, with hanging indent and double-spaced entries as required by the Hassan Usman Katsina Polytechnic project proposal guidelines.

- Adeyemi, A. T., Bello, K. O., & Okoro, C. (2021). Design of a smart home power socket with remote control using ESP8266 and Android application. *Nigerian Journal of Technology*, 40(3), 412–420.
- Adewale, A. A., Okafor, K. C., & Onuoha, O. (2023). Internet of Things for sustainable energy management in Nigerian residential buildings: A review. *Journal of Energy Technology and Policy*, 13(2), 56–70.
- Babangida, M., Idris, A., & Yusuf, L. (2024). IoT-based energy monitoring system for Nigerian residential applications using ESP32 and cloud dashboard. *International Journal of Electrical and Computer Engineering*, 14(1), 145–158.
- Google. (2024). Guide to app architecture. Android Developers. Retrieved from <https://developer.android.com/topic/architecture>
- Google. (2025). Antigravity: AI-assisted development for Android. Google Developers Blog. Retrieved from <https://developers.googleblog.com/antigravity>
- Kumar, R., & Sharma, P. (2025). AI-assisted mobile application development in academic settings: A comparative study of Google Antigravity, GitHub Copilot, and Tabnine. *IEEE Transactions on Education*, 68(1), 45–53.
- Martinez, J. R., & Lopez, F. A. (2023). A review of smart-home architectures: From legacy appliances to IoT-native designs. *IEEE Communications Surveys & Tutorials*, 25(2), 1234–1262.
- Mohan, N., Undeland, T. M., & Robbins, W. P. (2023). *Power electronics: Converters, applications, and design* (4th ed.). John Wiley & Sons.
- Nigerian Communications Commission. (2024). Subscriber statistics report, December 2024. Abuja, Nigeria: NCC. Retrieved from <https://www.ncc.gov.ng/statistics>
- Oluwafemi, O., Adebayo, S., & Mohammed, D. (2023). Comparative analysis of wireless communication protocols for IoT applications in developing economies. *Journal of Network and Computer Applications*, 215, 103624.
- Park, S., Kim, J., Lee, H., & Choi, Y. (2022). Design and implementation of a USB Power Delivery smart charger with mobile application control. *IEEE Access*, 10, 42105–42116.
- Rashid, M. H. (2022). *Power electronic devices, circuits, and applications* (5th ed.). Upper Saddle River, NJ: Pearson Education.
- Senthilkumar, R., Anitha, K., & Prabhu, M. (2024). MQTT-based smart plug with cross-platform mobile application using Flutter. *International Journal of Internet of Things and Web Services*, 9, 22–34.

- Snyder, H. (2020). Literature review as a research methodology: An overview and guidelines. *Journal of Business Research*, 104, 333–339.
- STMicroelectronics. (2023). LM317 adjustable linear voltage regulator datasheet (Rev. 12). Geneva, Switzerland: STMicroelectronics. Retrieved from <https://www.st.com/resource/en/datasheet/lm317.pdf>
- Texas Instruments. (2023). LM317 3-pin adjustable regulator datasheet (SNOSAV2D). Dallas, TX: Texas Instruments. Retrieved from <https://www.ti.com/lit/ds/symlink/lm317.pdf>
- Theraja, B. L., & Theraja, A. K. (2023). *A textbook of electrical technology* (Vol. 1, 25th ed.). New Delhi, India: S. Chand & Company.
- USB Implementers Forum. (2024). USB 3.2 and USB Power Delivery 3.1 specifications. Beaverton, OR: USB-IF. Retrieved from <https://www.usb.org/documents>
- Vincent, I. O., Okonkwo, E. A., & Bello, H. (2023). Reproducibility in undergraduate engineering research: Principles and practices. *Journal of Engineering Education Transformations*, 36(3), 1–16.